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* الكود (قد تمت مراجعة بيانات الطالب ورقم الجلوس)

ROIQ

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8

Which of the following is present in both HTTP request line and status line?

- A) HTTP version number
- B) Request method
- C) Status code
- D) URL

☒ A

☐ B

☐ C

☐ D

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9

Which of the following statements is true about circuit switching?

- A) Resources are pre-allocated regardless of demand
- B) Packet transmission can use the full link bandwidth
- C) The path between a source and a destination is not fixed
- D) All of the options

☒ A

☐ B

☐ C

☐ D

 **Will be reviewed**

10

Which of the following statements is true about TCP and not true about UDP?

- A) Provides timing service.
- B) Provides minimum throughput guarantee service.
- C) Provides security service.
- D) Provides reliable transport service.

☐ A

☐ B

☐ C

☒ D

 **Will be reviewed**

11

A Web server stores a webpage that comprises a base HTML file and 2 images referenced by the base HTML file. The HTML file is 100 bytes and each image is 200 bytes. A client is connected to the Web server through a direct link of 1 Mbps. Propagation delay between the Web server and the client is 50 milliseconds.

The client downloads the webpage using persistent HTTP but without pipelining (i.e. the next HTTP request is sent after the response for the previous HTTP request is received). Assume HTTP header, TCP header and ACK packets are of negligible size, transmission channel is perfectly reliable, time to establish and close TCP connection can be ignored. How long (in milliseconds) does it take for the client to download the entire webpage from the Web server?

A) 404

- B) 604
- C) 204
- D) 402.4

☒ A

☐ B

☐ C

☐ D

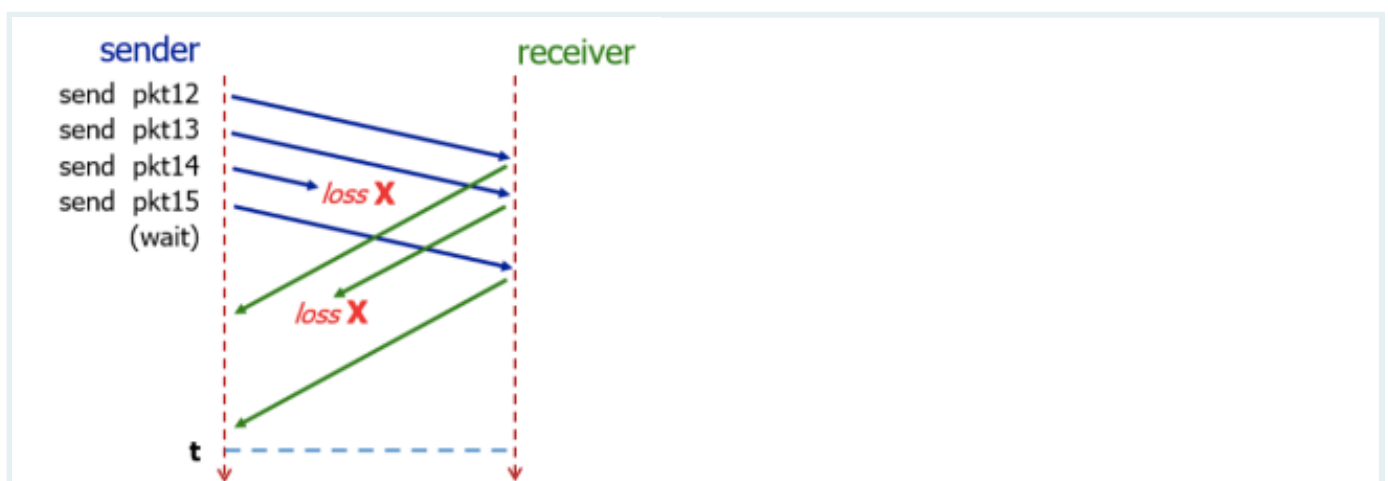
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12

Consider a sender and a receiver communicating using Selective Repeat protocol. Every packet embeds a 4-bit sequence number field. Sender window size is 4. None of the packets shown in the following figure are corrupted packets. However, the third data packet and the second ACK packet are lost.

What is the sender window over the sequence number space at time t ?

- A) 12, 13, 14, 15
- B) 13, 14, 0, 1
- C) 13, 14, 15, 0
- D) 13, 14, 15, 16



☐ A

☐ B

☒ C

☐ D

 **Will be reviewed**

13

Which of the following statements is true?

- A) Both the Internet and traditional telephone networks use packet-switching.
- B) Performance metrics such as end-to-end delay and throughput can be guaranteed in the Internet, if TCP is chosen as the transport layer protocol.
- C) The Internet uses packet switching and traditional telephone networks uses circuit switching.
- D) In the Internet, packets from the same source always take the same path to reach destination.

☐ A

☐ B

☒ C

☐ D

 **Will be reviewed**

14

Which of the following statements about DNS is TRUE?

- A) TLD servers store all the hostname to IP mappings of the Internet.
- B) Every Web server must have a canonical name.
- C) Local DNS server sometimes may provide out-of-date hostname-to-IP-address mapping.
- D) DNS server listens to TCP port 53.

☐ A

☐ B

☒ C

☐ D

 **Will be reviewed**

15

How many sockets are there in a TCP server communicating with 13 clients concurrently?

- A) 13
- B) 14
- C) 15
- D) 12

☐ A

☒ B

☐ C

☐ D

 **Will be reviewed**

16

A source and a destination are separated by 3×10^5 kilometers and are connected by a direct link of 2 Kbps. The propagation speed over the link is 2×10^8 meters/second. The source sends 100 packets to the destination using RDT 2.2. Each packet is of 2×10^3 bits long.

Suppose ACK packets are of negligible size and transmission channel is perfectly reliable. What is the throughput (in bps) of transmission?

- A) 200
- B) 500
- C) 100
- D) 400

☐ A

☒ B

☐ C

☐ D

📝 Will be reviewed

17

Which of the following statements is true about packet switching?

- A) In packet switching, bandwidth is divided into “pieces” between the nodes.
- B) In packet switching, dedicated allocation is done between nodes.
- C) In packet switching, resources are reserved for nodes.
- D) None of the options.

☐ A

☐ B

☐ C

☒ D

📝 Will be reviewed

18

Consider the following Python code snippet.

Suppose no exception is raised when the above code snippet is executed, which of the following statements is TRUE?

- A) s1 is a UDP socket
- B) Server uses s1 to transmit application data to client
- C) s2 listens to port 8080
- D) s2 listens to a random port number assigned by operation system

```
...  
s1.bind(('', 8080))  
s2, addr = s1.accept()  
...
```

☐ A

☐ B

☒ C

☐ D

 Will be reviewed

19

A browser requests to download index.html from a Web server. The HTTP response header received by the browser is shown below.

Which of the following statements is TRUE?

- A) In the HTTP request, browser has requested for a non-persistent connection.
- B) Suppose TCP header is 20 bytes, the length of the TCP segment containing the HTTP response is 1276 bytes.
- C) The index.html file received by the browser may have been corrupted during transmission.
- D) The Web server uses a cookie to keep the state information of the client.

```
HTTP/1.1 200 OK
Accept-Ranges: bytes
Content-Type: text/html; charset=UTF-8
Date: Sun, 20 Sep 2020 12:30:48 GMT
Expires: Sun, 27 Sep 2020 12:30:48 GMT
Last-Modified: Thu, 17 Oct 2019 07:18:26 GMT
Set-Cookie: PHPSESSID=6puvcfqiu7f365rf7vc6ud7r44;
path=/
Content-Length: 1256
```

☐ A

☐ B

☐ C

☒ D

 Will be reviewed

20

In circuit switching, circuit segment is idle if not used by call.

- A) True
- B) False

☒ A

☐ B

📝 Will be reviewed

21

Which of the following layers doesn't appear in the Internet Protocol Stack?

- A) Application layer
- B) Network layer
- C) Transport layer
- D) Session layer

☐ A

☐ B

☐ C

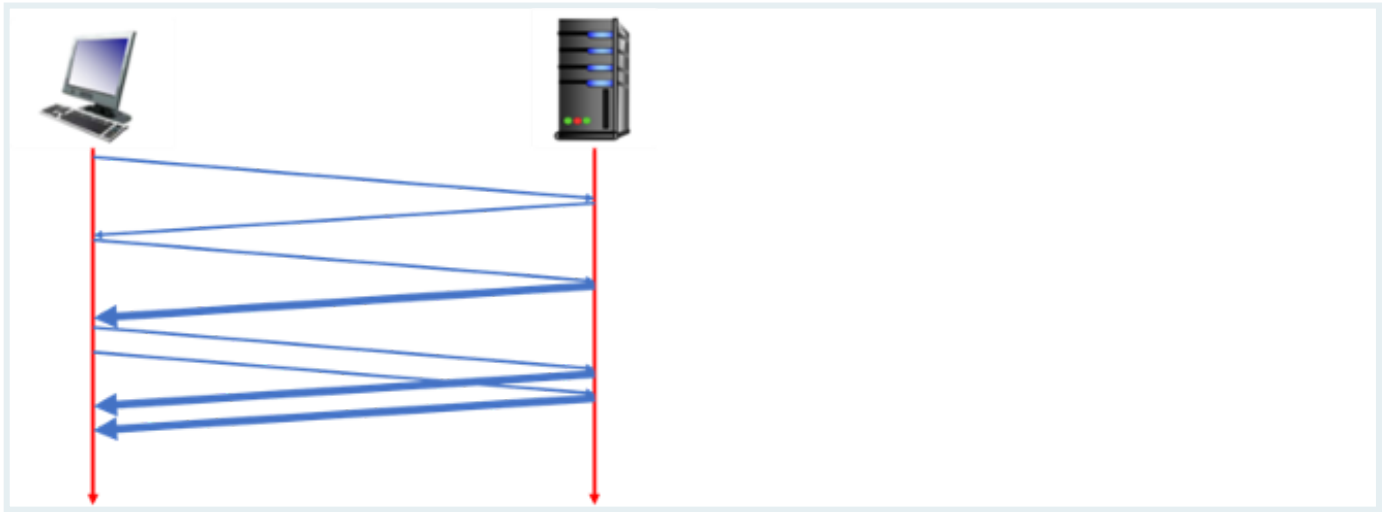
☒ D

📝 Will be reviewed

22

Which of the following HTTP protocols does the given figure represent to fetch a web page with 2 reference objects?

- A) Non-persistent HTTP.
- B) Persistent HTTP.
- C) Non-persistent HTTP with parallel connection.
- D) Persistent HTTP with pipelining.



- ☐ A
- ☐ B
- ☐ C
- ☒ D

📝 Will be reviewed

23

Can HTTP response message contains an empty body? If yes, When can this happen?

- A) No, an HTTP response message can never have an empty body.
- B) Yes, if the requested object has been moved to a new location.
- C) Yes, if the requested object is recently modified.
- D) Yes, if the requested object is very small in size.

- ☐ A
- ☒ B
- ☐ C
- ☐ D

📝 Will be reviewed

A packet switch receives a packet and determines the outbound link to which the packet should be forwarded. When the packet arrives, x bits of the currently-being-transmitted packet have been transmitted on this outbound link and n other packets are waiting to be transmitted. Packets are transmitted continuously in order of arrival. Suppose all packets are L bits and the link rate is R bps.

What is the queuing delay?

- A) nL / R
- B) $(nL + (x - L)) / R$
- C) $n(L - x) / R$
- D) $(nL + (L - x)) / R$

☐ A

☐ B

☐ C

☒ D

 **Will be reviewed**

In Go-back-N protocol, what are the size of the respective sender and receiver buffers required for a window size of N ?

- A) 1; 1
- B) $N - 1$; 1
- C) $N - 1$; $N - 1$
- D) N ; 1

☐ A

☐ B

☐ C

☒ D

 **Will be reviewed**

26

Using Conditional GET in HTTP Protocol, the number of RTT is reduced if the object is not modified since the date of the cached copy!

- A) True
- B) False

☐ A

☒ B

 **Will be reviewed**

27

Which of the following statements is true about Client-Server Architecture?

- A) In Client-Server Architecture, the server waits for incoming requests while the client Initiates contact with server.
- B) In Client-Server Architecture, the client waits for incoming requests while the server Initiates contact with server.
- C) In Client-Server Architecture, the server typically requests service from the client while the client provides requested service to server.
- D) All of the options.

☒ A

☐ B

☐ C

☐ D

 **Will be reviewed**

28

HTTP protocol keeps state information at the server side about past client requests.

- A) True
- B) False

☐ A

☒ B

 **Will be reviewed**

29

Consider the transmission between a UDP sender and a UDP receiver. Which of the following will never happen? You may assume that the application riding on UDP doesn't implement any reliability mechanisms.

- A) UDP receiver fails to receive any packet from UDP sender.
- B) UDP receiver receives out-of-order packets from UDP sender.
- C) UDP receiver receives duplicate packets from UDP sender.
- D) UDP receiver receives corrupted packets from UDP sender but fails to detect bit errors.

☐ A

☐ B

☒ C

☐ D

 **Will be reviewed**

30

End-to-end delay is the time taken for a packet to travel from source to destination. It consists of which of the following delays?

- A) Transmission delay.
- B) Propagation delay.

- C) Queuing and Processing delays.
- D) All of the options

☐ A

☐ B

☐ C

☒ D

 **Will be reviewed**

31

In Non-persistent HTTP, the client closes the TCP connection after fetching one object (one pair of HTTP request and HTTP response)

- A) True
- B) False

☒ A

☐ B

 **Will be reviewed**

32

A Selective Repeat sender just receives an ACK packet with ACK number 7. This ACK number falls within sender window which has the window size 3. Every data packet embeds a k-bit sequence number field (k is a constant unknown to you). Which of the following definitely CANNOT be the sequence number of the next packet transmitted by the sender?

- A) 0
- B) 2
- C) 3
- D) 6

☐ A

☐ B

☒ C

☐ D

 **Will be reviewed**

33

Assume a sender and a receiver connected via one packet-switch. The sender sends a message of size 24 KBytes using packets of length 1000 Bytes. The bit rates are 2 kbps and 5 kbps for the first and the second link, respectively. Using store and forward, what is the end-to-end delay (in seconds)? You may ignore the propagation delay!

- A) 16.8
- B) 134.4
- C) 0.1008
- D) 16800

☐ A

☒ B

☐ C

☐ D

 **Will be reviewed**

34

Top-level domain (TLD) servers are organization's own DNS server(s), providing authoritative hostname to IP mappings for organization's named hosts.

- A) Yes
- B) No

☐ A



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